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*Answer* **all** *questions in the spaces provided.*

**1.** There are different types of material in use today. Their uses are determined by their properties.

*(a)* (i) Complete the table below by selecting the word from the box to identify the type of material from its description.

*Each word may be used once, more than once or not at all.* [3]

| | | | | |
|---|---|---|---|---|
| polymer | ceramic | composite | alloy | metal |

| Description | Type of material |
|---|---|
| hard,<br>brittle,<br>high resistance | .................................... |
| hard,<br>high melting point,<br>good conductor | .................................... |
| mixture of two or<br>more elements | .................................... |

(ii) The properties of materials depend on their structure.
Describe the structure of polymers. [2]

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*(b)* Metals are very useful materials in a variety of industries. One use is for making wings of modern aircraft.

Some properties of the metals that could be used are given in the table below.

| Metal | Density ($kg/m^3$) | Stiffness (GPa) | Tensile strength ($\times 10^7$ Pa) |
|---|---|---|---|
| titanium | 4500 | 110 | 55 |
| aluminium | 2700 | 69 | 10 |
| vanadium | 5700 | 138 | 63 |
| steel | 7800 | 210 | 40 |

Use the information from the table above to answer the following questions.

(i) State which metal could be used to make the wings as flexible as possible. [1]

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(ii) I State which metal could be used to make the wings as light as possible. [1]

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II Give **one** reason for your answer. [1]

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(iii) I State which metal could be used so the wings can withstand large forces. [1]

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II Give **one** reason for your answer. [1]

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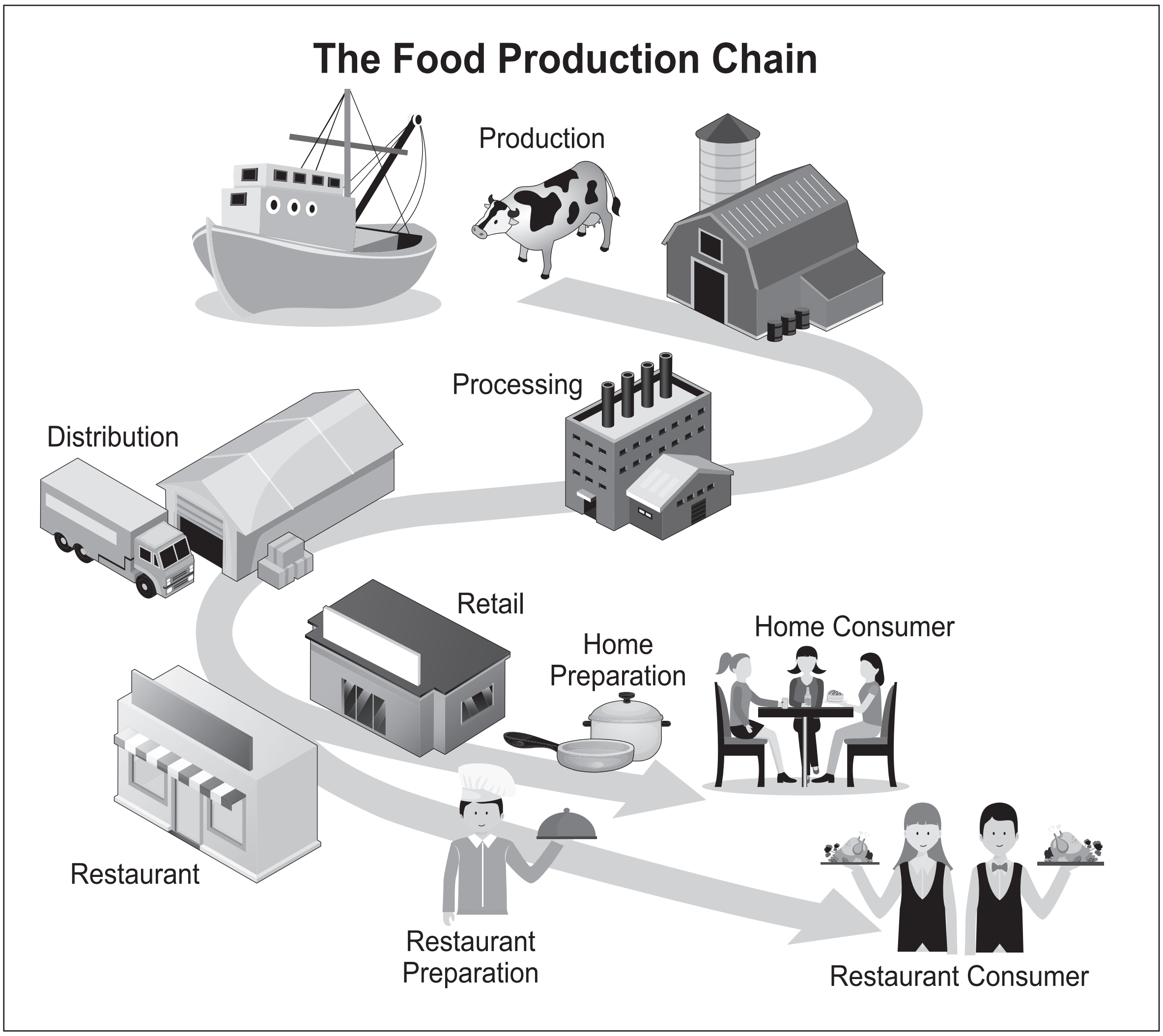
**2.** The food industry uses fruits, vegetables, grains and raw meat and processes these into the food products that are sold in supermarkets. Food production often involves cleaning and cooking raw foods, as well as the addition of additives and other ingredients.

One purpose of food production processes is to create products with longer shelf-lives than raw food ingredients.

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The food industry must take precautions to prevent consumers becoming ill from food poisoning.

(a) Complete the table below which shows some of the methods of controlling bacteria in the food industry. [3]

| Method | Effect on bacteria |
|---|---|
| refrigeration | slows down bacterial growth |
| freezing | .................................................... |
| .................................... | kills bacteria |
| .................................... | dries bacteria to prevent growth |

*(b)* Sometimes food becomes contaminated with bacteria, causing food poisoning.

(i) State **one** symptom of food poisoning. [1]

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(ii) Name **one** type of bacteria that causes food poisoning. [1]

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(iii) State the harmful substances produced by bacteria as they grow. [1]

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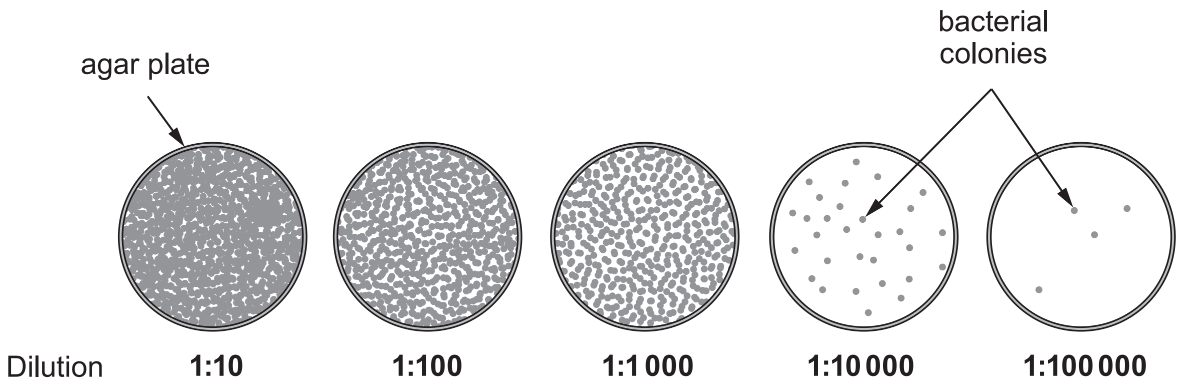
*(c)* Whenever an outbreak of food poisoning occurs, environmental health officers test samples of food for the presence of bacteria.

(i) To be able to count the number of bacterial colonies a series of dilutions is carried out as shown in the diagram below.




The 1:10 000 dilution was used to calculate the number of bacteria present.

I Explain why using a dilution of 1:10 will not provide valid results. [2]

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II Give **one** reason why the 1:100 000 dilution was also not used to calculate the number of bacteria present. [1]

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(ii) In an investigation, six agar plates were prepared by adding 1 $cm^3$ of solution from a 1:10 000 dilution. The agar plates were incubated for 72 hours. The results are shown below.

| Plate number | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Number of colonies | 35 | 35 | 25 | 15 | 30 | 25 |

I State which plate has produced an anomalous result. [1]

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II Use the equation:

number of bacteria/$cm^3$ = mean number of colonies × 10 000

to calculate the number of bacteria/$cm^3$ in the original sample. [3]

Number of bacteria/$cm^3$ = .................................................

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**3.** Forensic scientists analyse substances found at crime scenes. They routinely identify powder samples.

Students are using the same techniques to analyse an unknown powder.

*(a)* The students dissolve the unknown powder in water and add sodium hydroxide solution (NaOH) to test for metal cations. They then add excess NaOH solution to see if the precipitate dissolves or not.

The table shows results for a selection of cations.

| Cation | Colour of precipitate when a little NaOH solution is added | Precipitate dissolves when excess NaOH solution is added (✓ or ×) |
|---|---|---|
| $Ca^{2+}$ | white | × |
| $Zn^{2+}$ | white | ✓ |
| $Pb^{2+}$ | white | ✓ |
| $Mg^{2+}$ | white | × |
| $Cu^{2+}$ | blue | × |
| $Fe^{2+}$ | green | × |

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They noticed that a white precipitate was produced which did not dissolve in excess NaOH solution.

(i) State the symbols for the **two** metal cations that could be present. [1]

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(ii) The table below shows the flame tests for a selection of metal cations.

| Metal cation | Flame colour |
|---|---|
| $Ca^{2+}$ | brick-red |
| $Zn^{2+}$ | greyish-white |
| $Pb^{2+}$ | greyish-white |
| $Mg^{2+}$ | no colour |
| $Cu^{2+}$ | blue-green |
| $Fe^{2+}$ | gold |

Describe how a flame test could be used to confirm which cation is present. [2]

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*(b)* They use another sample of the powder and perform the following tests to identify the anions that may be present.

| Anion | Test | Observation |
|---|---|---|
| bromide ($Br^-$) | add aqueous silver nitrate | cream precipitate |
| carbonate ($CO_3^{2-}$) | add hydrochloric acid | bubbles of gas |
| chloride ($Cl^-$) | add aqueous silver nitrate | white precipitate |
| iodide ($I^-$) | add aqueous silver nitrate | yellow precipitate |
| sulfate ($SO_4^{2-}$) | add aqueous barium chloride | white precipitate |

The observations from the students' tests are:

- bubbles of gas are given off when hydrochloric acid is added
- a white precipitate forms when aqueous silver nitrate is added

Use the information above to determine the **two** anions present in the powder. [2]

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*(c)* Name **one** compound that could be present in the powder. [1]

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